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Digital twin-based multiscale fatigue crack life prediction of rib-to-deck weld joints

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ABSTRACT

Fatigue cracks are prone to initiate in rib-to-deck (RTD) weld joints of orthotropic steel decks (OSD) under traffic loading, posing a serious threat to functionality and safety of bridge structures. To achieve accurate fatigue crack life prediction of RTD weld joints of OSD, a digital twin-based multiscale fatigue crack life prediction framework from microcrack initiation and propagation to macrocrack propagation is proposed. Eccentric tensile fatigue tests as a physical entity are first conducted to obtain test data on fatigue crack propagation. A multiscale crack propagation model coupling crystal plasticity finite element model with fracture mechanics is then established as a virtual entity to describe microcrack initiation and propagation as well as macrocrack propagation. A dynamic Bayesian network integrated with a particle filter algorithm is finally introduced to fuse the test data from the physical entity with the numerical results from the virtual entity, enabling real-time updating of key model parameters and crack propagation states as well as quantitative uncertainty assessment and digital twin construction. The results from this study demonstrate that the proposed digital twin effectively suppresses model uncertainty and significantly improves the prediction accuracy of crack propagation process and remaining fatigue life.

1. Introduction

Orthotropic steel decks (OSD) have been widely adopted in long-span bridges owing to their favorable mechanical performance. Nevertheless, under long-term traffic loading, rib-to-deck (RTD) weld joints of OSD are highly susceptible to fatigue cracks and compromise the functionality and safety of bridge structures [1]. Therefore, an accurate fatigue life prediction of RTD weld joints of OSD is of great significance, as it will facilitate the early identification of potential fatigue damage and provide a scientific basis for condition assessment and maintenance strategies during the bridge operation.

The fatigue life of RTD weld joints can generally be divided into the stages of microcrack initiation, microcrack propagation, and macrocrack propagation. Among these stages, the microcrack initiation and propagation usually account for a dominant proportion of the total fatigue life. The microcrack initiation and propagation are primarily determined by the microstructural characteristics of the material. The welded joints are heterogeneous, consisting of weld metal, heat-affected zone (HAZ), and base metal. Previous studies have shown that both weld metal and base metal can exhibit significant cyclic plasticity under fatigue loading conditions [2]. The HAZ has been widely reported as a preferential region for plastic strain concentration [3]. Even when the nominal stress is below the macroscopic yield strength of the material, fatigue crack initiation is primarily governed by localized cyclic plastic deformation at stress concentration regions rather than uniform plastic deformation, as local stress-strain incompatibilities at geometric discontinuities or stress concentration regions can induce such localized deformation, promoting the formation of slip bands and initiating fatigue microcracks [4]. Under cyclic loading, the welded joints exhibit typical mixed-mode crack characteristics, in which both normal and shear stress components act simultaneously at the crack tip. The transition from microcrack initiation and propagation to macrocrack propagation is associated with the evolution of the local crack tip stress field and the progressive accumulation of damage [5]. During cyclic loading, microcracks propagate and coalesce. Once the local stress intensity factor or energy release rate reaches critical threshold, crack propagation proceeds in accordance with fracture mechanics principles [6].

As a powerful analytical tool, the finite element method is capable of

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simulating crack initiation and propagation of steel bridge structures [7,8]. At the microscale, the material in the weld zone is significantly affected by welding heat input, resulting in complex grain size, morphology, and orientation distributions which play a critical role in the microcrack initiation and propagation [9]. Zhou et al. [10] performed numerical simulation of fatigue microcrack initiation and propagation in weld zone based on crystal plasticity finite element model (CPFEM) implemented in ABAQUS. CPFEM explicitly accounts for grain randomness, thereby enabling effective simulation of the microscale mechanical response of materials under cyclic loading and capturing localized stress-strain states. At the macroscale, the S-N curve method is a commonly used approach for fatigue life assessment but cannot be used to simulate the crack propagation process. The fracture mechanics-based methods, and damage mechanics-based methods are often used to estimate macrocrack propagation [11]. Among these, fracture mechanics-based methods can describe crack propagation behavior with relatively higher efficiency and accuracy than other two methods. Luo et al. [12] combined ABAQUS and Franc3D to establish a macrocrack propagation model for steel bridges based on fracture mechanics theory, through which the propagation behavior of macrocrack in RTD weld joints were systematically investigated.

However, the application of CPFEM in numerical simulations requires solving complex nonlinear constitutive relations at each integration point, which leads to a high computational cost. Consequently, this method is generally applicable only to the stages of microcrack initiation and early propagation, and it is difficult to simulate crack propagation at macroscale. In contrast, the fracture mechanics-based methods have difficulty in accurately characterizing the mechanical behavior of materials subjected to pronounced plastic deformation. The predictions are highly sensitive to the assumed initial crack size, thereby limiting their predictive capability during the microcrack propagation stage. Despite these limitations, fracture mechanics-based methods can simplify complex crack propagation phenomena into relationships determined by a limited number of key model parameters, thereby enabling the computationally efficient modeling of macrocrack propagation. Therefore, it is necessary to establish a multiscale numerical model to achieve accurate prediction of the entire fatigue crack propagation process from microcrack initiation and propagation to macrocrack propagation while maintaining computational efficiency.

It is also noted that in the prediction of fatigue crack propagation process using either CPFEM or fracture-mechanics-based methods, the results always exhibit a strong dependency on key model parameters. On the other hand, adopting a fixed set of key model parameters without dynamic updating often fails to fully capture the inherent randomness and uncertainties of crack propagation observed during fatigue tests. As a key emerging technology, digital twin (DT) is capable of efficiently fusing multi-source test data with high-fidelity numerical simulation models, characterized by continuous evolution and adaptive updating capabilities. Owing to its advantages in real-time perception, online parameter updating, and life-cycle management, DT provides a novel technical pathway for the intelligent operation and maintenance of bridge structures [13]. Specifically, Zeng et al. [14] proposed a sequential DT framework that integrates corrosion information and load data of concrete beams to construct multi-level DTs for parameter optimization, achieving high prediction accuracy. Lao et al. [15] utilized DT to fuse microcrack characterization data with CPFEM to dynamically eliminate model uncertainties, thereby realizing the fatigue life prediction of weld microcracks. Cheok et al. [16] established a DT based on strain interfaces, combining Bayesian inference and Gaussian process regression to achieve real-time diagnosis of weld crack propagation and remaining life prediction. Peng et al. [17] developed a data-driven DT method using dynamic Bayesian networks(DBN) for online updating of crack propagation parameters, which, coupled with the weight function method, enabled high-accuracy fatigue life prediction.

This study therefore proposes a digital-twin based multiscale numerical model to achieve accurate prediction of the entire fatigue crack propagation process from microcrack initiation and propagation to macrocrack propagation of RTD weld joints while maintaining computational efficiency. The remainder of this paper is organized as follows. Section 2 presents the overall framework of the digital twin-based fatigue crack propagation and life prediction approach. Section 3 briefly introduces the design of eccentric tensile specimens and describes the corresponding fatigue test program and the obtained test data. Section 4 details the development of the virtual entity, which is established based on a multiscale crack propagation model combining microcrack and macrocrack propagation models. Section 5 elaborates on the construction of the DT using a DBN, where test data and multiscale numerical simulation results are fused to update key model parameters. Section 6 presents the prediction results of crack propagation process and remaining fatigue life of the test specimens based on the established DT. Finally, Section 7 summarizes the main conclusions of this study.

2. A framework for digital twin-based approach

A DT normally consists of four interrelated components [18]: (a) a physical entity and its test data acquisition; (b) a virtual entity and its simulation data; (c) the DT construction through data fusion; and (d) the application and prediction of the DT. Specifically, the DT proposed in this study integrates fatigue test data and multiscale numerical simulation results within a DBN framework, by which fatigue crack states are inferred, estimated, and updated online and thereby the real-time updating of the entire fatigue crack propagation and the probabilistic prediction of remaining fatigue life are realized. The overall framework of the proposed DT is illustrated in Fig. 1, and its key components are described as follows:

(a) Physical entity and test data acquisition: A novel eccentric tensile specimen is designed and fabricated to simulate the realistic stress state of RTD weld joints. Systematic fatigue tests are conducted with continuous monitoring of fatigue crack propagation process. Crack length is obtained and serves as reliable observation information for step (c).

(b) Virtual entity and simulation data: Based on the fatigue tests of the eccentric tensile specimens, a multiscale numerical model for crack propagation is established as a virtual entity. The virtual entity simulates the crack initiation and propagation process and provides simulation results, which include crack propagation paths, propagation rates, and stress–strain fields and serve as effective prior state information for step (c).

(c) DT construction and data fusion: Within the DBN framework, a particle filter (PF) algorithm is introduced to perform system inference and updating. Test data is fused with multiscale simulation results to achieve online estimation of state variables and key model parameters associated with fatigue crack initiation and propagation. Information exchange between the physical and virtual entities is realized by continuously updating the prior distributions of the parameters, enabling DT construction.

(d) Application and prediction: The established DT is employed to predict the microcrack initiation and propagation and macrocrack propagation processes and obtain the probabilistic distributions of remaining fatigue life of the RTD weld joints.

3. Physical entity and test data acquisition

The fatigue tests of eccentric tensile specimens recently conducted by the writers [19] are taken as the physical entity in this study. This section briefly introduces the design and fabrication of eccentric tensile fatigue specimens and describes the corresponding fatigue test program and the obtained test data.

3.1. Design of test specimens

To realistically simulate the combined tensile and bending stress

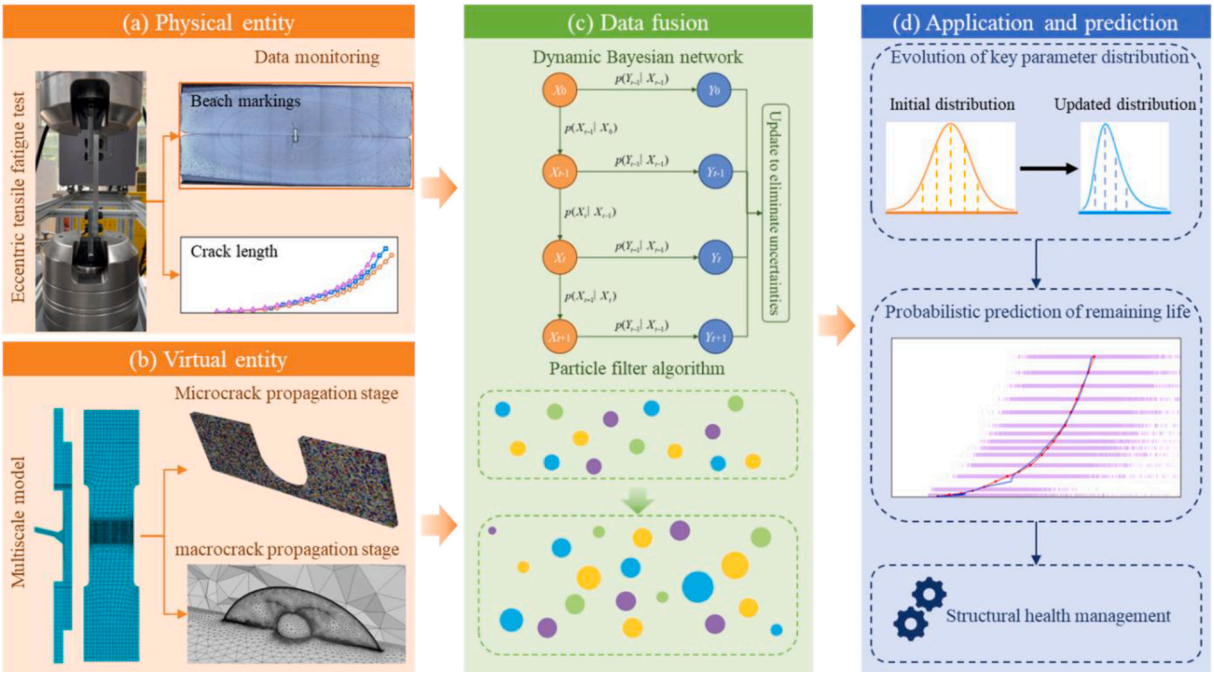


Fig. 1. Digital twin framework.

state of RTD weld joints in OSD under vehicle loading, a novel eccentric tensile specimen was designed, as shown in Fig. 2. The specimens were extracted from full-scale OSD segments fabricated using actual bridge construction techniques, in which both bridge deck and U-ribs were made of Q345qD bridge steel with a thickness of 16 mm and 8 mm, respectively. A CO_2 gas-shielded full-penetration double-sided welding process was employed. The filler material is HTW-58 (Φ1.2) welding wire. Table 1 and Table 2 list the chemical compositions and mechanical properties of the base metal and filler metal, respectively, in which f_y denotes the yield strength, f_u denotes the ultimate tensile strength, and δ denotes the elongation. The detailed welding parameters have been provided in Table 3.

The segments were sectioned to obtain test specimens with a working section dimension of 80 mm × 16 mm and a retained U-rib length of 50 mm. The gripping ends of the specimens were widened to 100 mm to mitigate end stress concentration, and four bolt holes were provided at each end for connection to rectangle eccentric loading plates. To control crack initiation location and quantify propagation process, a circular hole defect with a diameter of 2 mm and a depth of 2.15 mm was pre-fabricated at the center of the RTD weld joint of the test specimen in accordance with the weld quality criteria in EN ISO 5817 [20]. Additionally, the weld zone was finely ground to eliminate the effects of random initial defects, such as undercuts and slag inclusions, on crack initiation and propagation.

In order to observe the microstructure, the specimen surfaces were sequentially ground using abrasive papers ranging from 320# to 2000#,

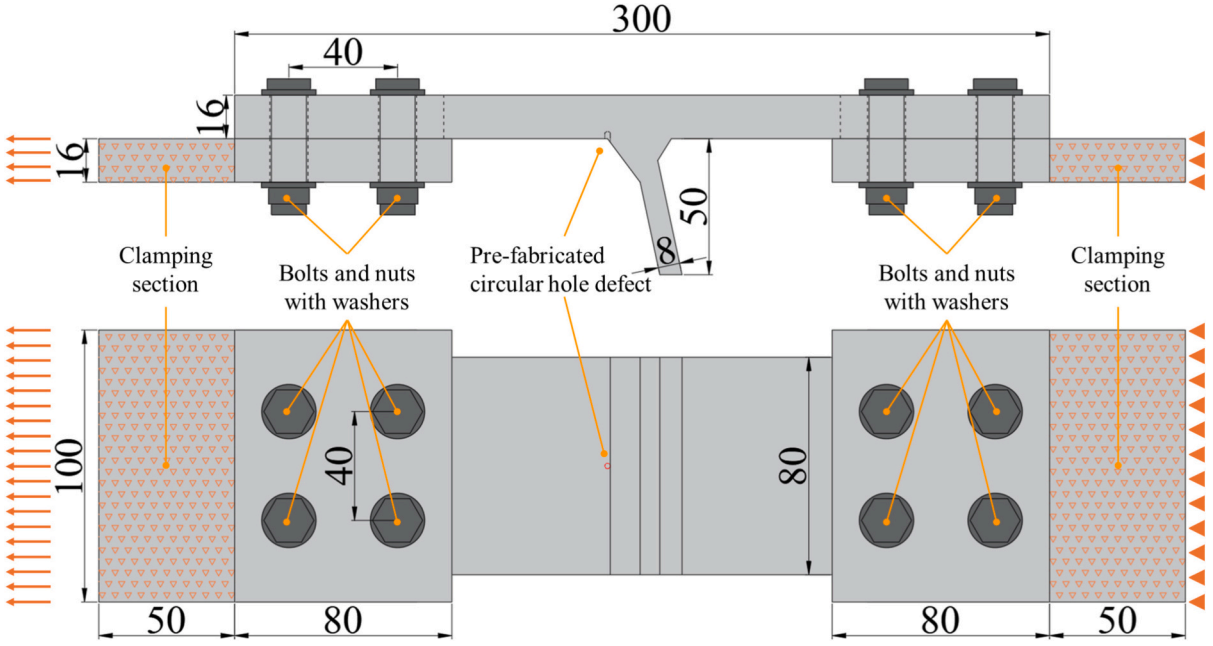


Fig. 2. Eccentric tensile fatigue test specimen (unit: mm) [19].

Table 1
Material chemical composition (%).

Material	C	Si	Mn	P	S	Als	Ni	Ti
Base metal	0.10	0.28	1.46	0.013	0.002	0.034	0.010	0.014
Filler metal	0.07	0.53	1.55	0.010	0.010	/	0.55	0.11

Table 2
Mechanical properties of Q345qD bridge steel.

Material	f_y(MPa)	f_u(MPa)	δ(%)	−40 °C(J)
Base metal	444	530	34	280,274,239
Filler metal	512	615	29	80,75,90

Table 3
Welding process parameters for test specimens.

Welding wire Material	Welding current (A)	Welding voltage (V)	Gas flow rate(L/min)
HTW-58(Φ1.2)	240 ± 20	30 ± 3	15–20

followed by mechanical polishing with polishing paste to remove surface impurities and oxide layers. The polished specimens were then chemically etched using a nitric acid solution to reveal the grain boundaries. After etching, the specimens were ultrasonically cleaned in alcohol to remove residual contaminants. The microstructures were observed using an optical microscope, and representative micrographs have been provided in Fig. 3(a). Subsequently, the acquired microstructure images were processed and analyzed using Image-Pro Plus software, which is widely used for optical micrograph processing and grain size measurement [21]. For the relatively complex microstructures considered in this study, automatic image preprocessing techniques were first applied to enhance image contrast, followed by manual marking to improve grain boundary identification accuracy, as shown in Fig. 3 (b)-(c). During the image analysis process, most grain boundaries were clearly identifiable; however, a small number of grain boundaries were partially obscured by impurities. Grains with indistinct boundaries were excluded from statistical measurements to minimize potential measurement errors. By statistically analyzing multiple optical micrographs, the grain size distribution data was finally obtained.

In this study, fatigue cracks initiate and propagate from the predefined circular hole defect as shown in Fig. 2, which is located within the HAZ. Therefore, both the metallographic characterization and the CPFEM modeling in this study focus on this specific region, thereby reducing the influence of large topological differences among different weld zones. In the established microcrack propagation model in next section, the depth of the polycrystal part used for the microcrack propagation simulation is 2.5 mm. This polycrystal part is located within the HAZ, where the thermal exposure during welding is considered to be comparable; therefore, the grain sizes are expected to be similar [22]. Consequently, although the internal subregions of the HAZ cannot be completely distinguished during specimen preparation, the material within the analyzed region can still be reasonably treated as belonging to a statistically homogeneous HAZ. The measured grain size distribution within the HAZ is presented in Fig. 4. Statistical analysis shows that the grain size distribution in this region can be well described using a lognormal distribution, with the functional form given in Eq. (1). Based on the equivalent circular diameter definition, the mean grain size is approximately 53.6 μm, with logarithmic mean μ and standard deviation σ values of 4.0 and 0.3, respectively.

$$f(x;\mu,\sigma) = \frac{1}{x\sigma\sqrt{2\pi}}\exp\left[\frac{-(\ln(x)-\mu)^2}{2\sigma^2}\right] \quad (1)$$

3.2. Fatigue tests

Two loading conditions were considered in this study with a nominal loading stress amplitude of 50 MPa and 60 MPa, respectively, and three fatigue test specimens were used for each loading condition. The fatigue tests were performed using a WANCE HDT255B electro-hydraulic servo control fatigue testing machine. Eccentric tensile loading was applied through rigidly connected rectangle steel plates at both ends of the specimen, thereby subjecting the RTD weld joint to a combined tensile-bending stress state. The test setup is shown in Fig. 5. To accurately measure crack length, the beach marking technique was employed to record the fatigue crack propagation path during the fatigue tests [23]. Since the loading amplitude used for beach marking differs from that of the fatigue loading (in this study, the stress ratio R of the fatigue loading was set to 0.1, while that of the beach marking loading R was set to 0.5), different energy inputs are introduced during the loading stages [24]. This results in variations in crack propagation rates, producing distinct propagation fronts along the crack path. These fronts appear as alternating bright and dark bands on the fracture surface (i.e., beach marks) [25], which visually represent the crack propagation trajectory. Prior to the formal tests, preliminary tests were conducted to determine suitable

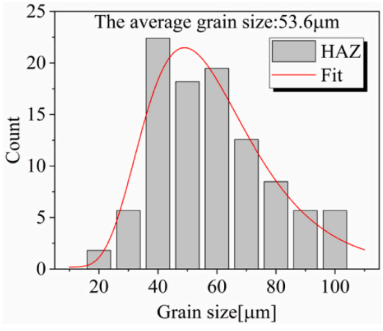


Fig. 4. Grain size distribution.

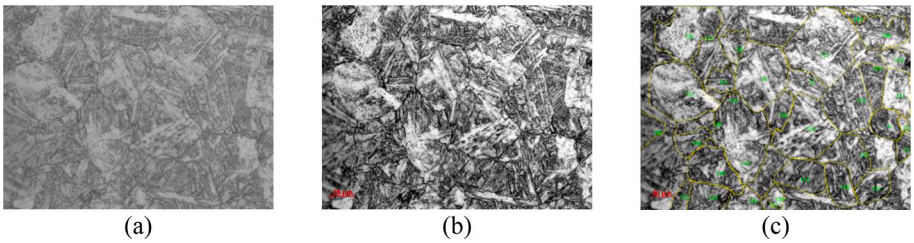


Fig. 3. Statistics of the grain size distribution: (a) optical micrograph, (b) preprocessed micrograph, (c) annotation of grain boundaries.

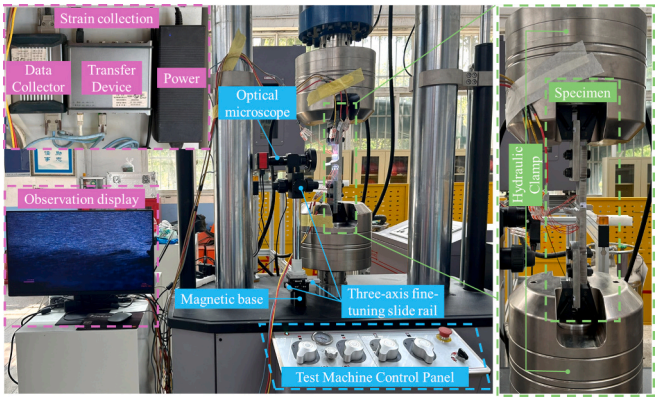


Fig. 5. Fatigue test arrangement [19].

beach marking loading parameters. 20,000 cycles were selected as the minimum number that could ensure clear beach marks without affecting normal crack propagation. Specifically, during fatigue testing, 20,000 cycles beach marking loading was applied after every 50,000 cycles in the early stage and every 100,000 cycles in the later stage. The fatigue crack propagation life was defined based only on the number of fatigue loading cycles, while beach marking cycles were used solely as auxiliary marks and were not included in fatigue life calculation. Previous studies have demonstrated that the beach marking method can produce clear and sufficient markings on fracture surfaces without interfering with normal crack propagation [26]. The test results indicate that when the number of beach marking cycles with the stress ratio of 0.5 is significantly lower than that of fatigue cycles with the stress ratio of 0.1, crack propagation during the beach marking stage is extremely limited, and its influence on the main crack propagation process can be neglected [27]. This loading produces clear and well-defined beach markings on the fracture surface without significantly interfering with the natural crack propagation behavior, thereby facilitating quantitative measurement of crack length using an optical microscope. In addition, prior to fatigue fracture of the specimen, the surface crack length was quantitatively measured using an optical microscope after each beach marking loading sequence. The detailed loading conditions are summarized in Table 4, and the loading sequence is illustrated in Fig. 6.

3.3. Test results

During the fatigue tests, surface cracks on the specimens were monitored using an optical microscope. After fatigue fracture, the fracture surface was imaged using an optical microscope, and the typical fracture surface morphology is shown in Fig. 7. The fracture surface can be clearly divided into three characteristic zones: the prefabricated crack zone, the stable crack propagation zone, and the final instantaneous brittle fracture zone. For all specimens, fatigue cracks were initiated from the prefabricated circular hole defect, with one microcrack forming on each side of the hole. Owing to factors such as machining tolerances of the prefabricated defect and deviations from ideal loading, variations were observed in the crack initiation positions, propagation paths, and final crack morphologies. The cracks propagated along the plate thickness direction, and an evident crack-front coalescence phenomenon was observed, eventually resulting in an approximately semi-elliptical crack shape [28]. When the crack depth reached approximately 10 mm, the specimen entered the stage of final instantaneous brittle fracture, during which the remaining fatigue life was significantly reduced, ultimately leading to brittle fracture.

The test results are summarized in Fig. 8. The crack length prior to crack coalescence is defined as the lengths of the two independent cracks, excluding the diameter of the prefabricated circular hole defect. After crack coalescence, the crack length is defined as the overall length of the merged crack formed after coalescence, which includes the diameter of the prefabricated circular hole defect. The crack coalescence refers to the process in which initially independent microcracks nucleated on both sides of the prefabricated hole propagate under cyclic loading. As the cracks propagate, their stress fields gradually overlap and interact, eventually merging into a larger main crack. This phenomenon typically occurs when adjacent cracks become sufficiently close so that the material ligament between them is eliminated, resulting in the formation of a unified crack front. It can be observed that both crack length and crack depth exhibit nonlinear propagation with increasing loading cycles. Under the nominal stress amplitude of 60 MPa, the crack propagation rate is significantly higher than that under 50 MPa. Meanwhile, a certain degree of scatter is observed in the crack propagation behavior among different specimens, which is mainly attributed to material microstructural heterogeneity, geometric tolerances of the prefabricated circular hole, and measurement uncertainties.

4. Virtual entity and multiscale simulations

The fatigue life of weld joints of OSD is considered in general to consist of two components: the microcrack initiation and propagation life N_{micro}, and the macrocrack propagation life N_{macro} [29].

$$N = N_{\text{micro}} + N_{\text{macro}} \tag{2}$$

In this study, the microcrack propagation life N_{micro} refers to the number of cycles required for the crack to propagate from the initiation length to the micro-macro transition size, while the macrocrack propagation life N_{macro} denotes the fatigue life interval during which the crack continues to propagate from the micro-macro transition size until final fracture of the specimen.

4.1. Multiscale simulations and virtual entity

Previous studies [30] indicate that the initiation and early propagation of microcrack accounts for a substantial portion of the total fatigue life. The microcrack fatigue life is mainly determined by the microstructural characteristics of the material. Crystal plasticity theory can effectively capture the heterogeneous deformation behavior of the material at grain scale, and it is therefore particularly well suited for analyzing microcrack initiation and propagation life. However, crystal plasticity constitutive models are highly complex, and constructing a crystal plasticity model for the entire specimen region would lead to prohibitively high computational costs, thereby rendering their direct

Table 4
Loading conditions in tests [19].

Specimen	Nominal stress amplitude	Fatigue load(kN)			Beach marking load(kN)		
		Peak load	Valley load	Stress ratio	Peak load	Valley load	Stress ratio
Pl-50-1							
Pl-50-2	50 MPa	71.11	7.11		71.11	35.55	
Pl-50-3				0.1			0.5
Pl-60-1							
Pl-60-2	60 MPa	85.33	8.53		85.33	42.67	
Pl-60-3							

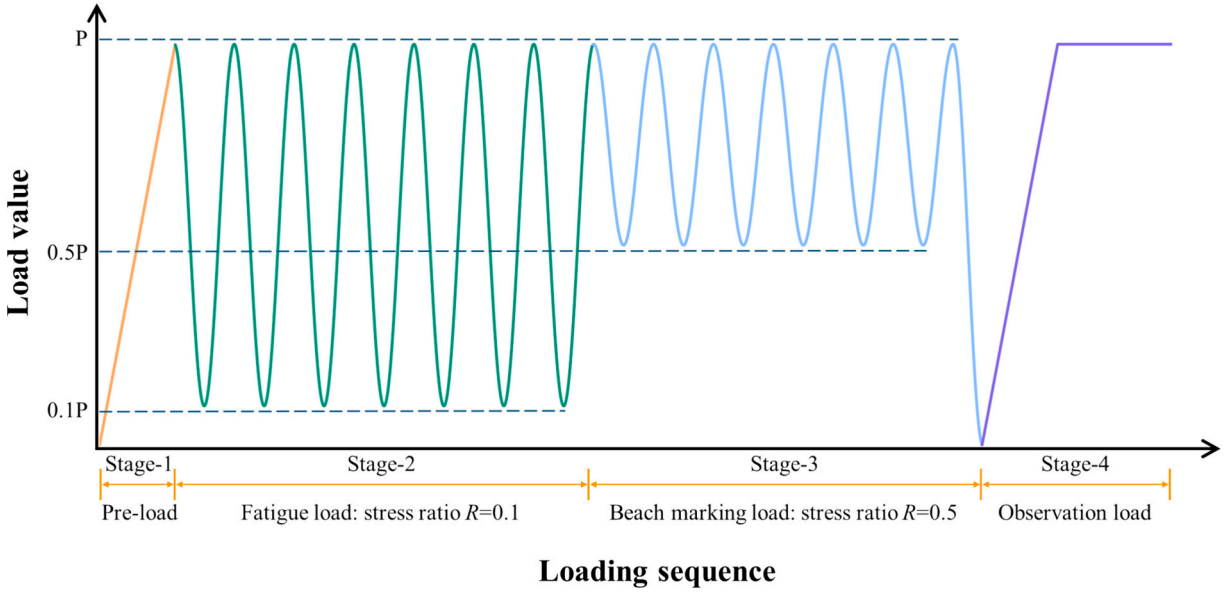


Fig. 6. Load sequences in tests [19].

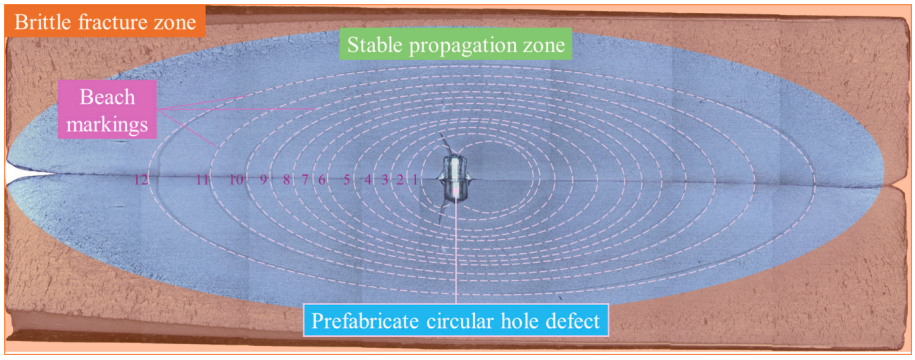


Fig. 7. Beach markings on fracture surface [19].

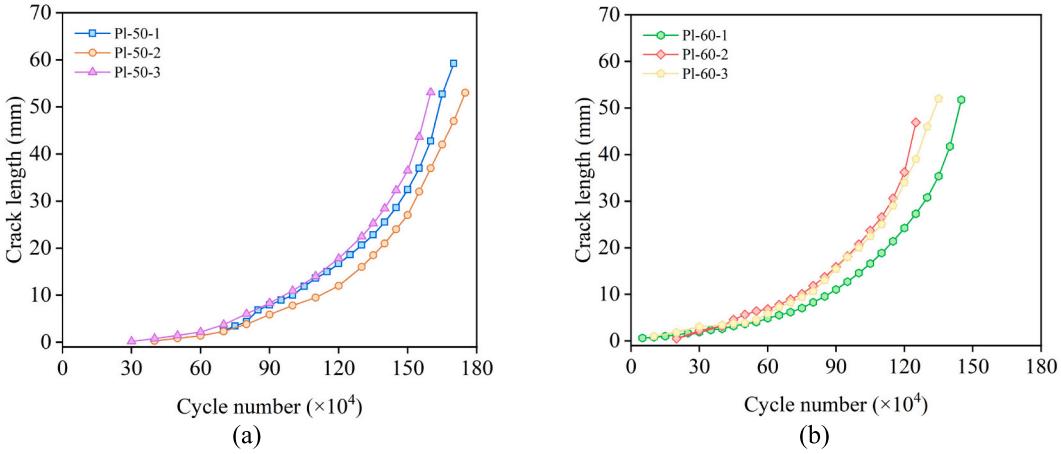


Fig. 8. Test results of crack length vs. cycle number: (a) 50 MPa; (b) 60 MPa [19].

application to the full crack propagation process impractical [31]. In contrast, linear elastic fracture mechanics (LEFM) is applicable to a wide range of geometries and loading conditions and can accurately characterize the stress and displacement fields near the crack tip by evaluating the stress intensity factor (SIF) along the crack front. Moreover, LEFM-based analyses only involve the solution of linear system equations, offering advantages such as fast convergence and low computational demand, which make them more suitable for efficient fatigue crack propagation simulation in the macrocrack propagation stage. Therefore, a multiscale crack initiation and propagation model is established in this

study as the virtual entity of the DT, with the overall flowchart of the proposed multiscale simulations illustrated in Fig. 9. Firstly, an initial finite element model is established in ABAQUS according to the geometric dimensions of the eccentric tensile specimen, with the prefabricated circular hole defect introduced in the weld. This model serves as the basic model for subsequent development of both the microcrack and macrocrack propagation models. It contains only geometric information and does not incorporate any specific constitutive equations. Owing to the fundamental differences in theoretical foundations and numerical methodologies between the microcrack and macrocrack propagation stages, separate numerical simulation strategies and constitutive equations are employed to establish a microcrack initiation and propagation model and a macrocrack propagation model, respectively. Specifically, CPFEM is employed to capture the evolution of microscale fatigue damage during the microcrack initiation and propagation stage, whereas LEFM is introduced to analyze crack propagation behavior in the macrocrack propagation stage.

4.2. Microcrack initiation and propagation stage

4.2.1. Fatigue life of microcrack initiation and propagation stage

For RTD weld joints subjected to cyclic loading, transgranular fracture is prone to occur within persistent slip bands inside individual grain. The activated slip planes gradually evolve into critical planes, leading to damage within the polycrystalline microstructure and ultimately triggering the microcrack initiation. It is often assumed that microcrack initiation occurs in the grain exhibiting the highest level of damage. With continued cyclic loading, the crack propagates along grain boundaries or active slip planes into adjacent grains, preferentially activating crystallographic slip systems in grains adjacent grain, and subsequently entering next grain until the crack penetrates the entire polycrystal part.

Based on the physical description of microcrack initiation and propagation, the corresponding crack propagation criteria and numerical simulation model can be established to quantitatively characterize the initiation and propagation behavior of microcracks within polycrystalline materials. The microcrack initiation and propagation in polycrystalline materials require appropriate damage criteria for effective control. Fatigue indicator parameter (FIP) has been widely employed in fatigue analysis of metallic materials [32–34]. In CPFEM-based fatigue life prediction, FIPs are commonly introduced using the plastic stress amplitude and plastic strain amplitude calculated on individual slip systems within grains, serving as criteria for microcrack initiation and propagation. The Gibbs free energy criterion proposed by Fine and Bhat [35] provides an effective framework for multiscale characterization of fatigue behavior in materials. At the microstructural scale, the accumulated distortion energy between grains is regarded as the dominant driving force controlling microcrack initiation and propagation. Under cyclic loading, persistent dislocation slip occurs along slip bands within crystals, while grain boundaries hinder dislocation motion, leading to gradual dislocation accumulation inside grains and the formation of localized energy concentrations. When the accumulated energy reaches a critical threshold comparable to the variation in Gibbs free energy, fatigue microcracks begin to initiate.

The maximum value of plastic strain energy density (PSED) ϖ of a single grain is used to determine the number of cycles required for fatigue crack initiation in this study. The microcrack fatigue life N_{micro} is considered as the sum of the crack initiation life of the initial grain and the microcrack propagation life through all subsequent grains, and it can be calculated by Eq. (3).

$$N_{\text{micro}} = 4\frac{\pi E\gamma_s - \sigma^2 d_0(1-\nu^2)}{\pi E f^* \varpi} + \sum_{i=1}^{n} 4\frac{\pi E\gamma_s - \sigma^2 d_i(1-\nu^2)}{\pi E f^* \varpi} \tag{3}$$

where γ_S is the surface energy of the material; σ is the nominal stress amplitude; ν is Poisson's ratio; E is Young's modulus; d_0 denotes the diameter of the initial grain; d_i represents the diameter of the ith grain and the grain size is statistically characterized by a lognormal distribution in this study; the parameter f^* is a key model parameter to be updated; $\varpi = \frac{\Delta\tau}{2}\frac{\Delta\gamma_p}{2}$ is PSED, which can be obtained from CPFEM simulations; $\Delta\tau$ is the plastic shear stress range; $\Delta\gamma_p$ is the plastic shear strain range.

4.2.2. Numerical simulation based on CPFEM

The modeling type of CPFEM generally includes 2D model and 3D representative volume element (RVE) model. Although 2D model offers relatively high computational efficiency, they are incapable of accurately representing the three-dimensional grain morphology, orientation distribution, and intergranular interactions. Moreover, simulations performed on isolated 3D RVEs are insufficient to capture the stress-strain states of specimens under realistic boundary conditions. Nevertheless, it has been demonstrated that embedding an RVE into a macroscale specimen model enables a more accurate representation of the local stress state of materials under realistic cyclic loading conditions

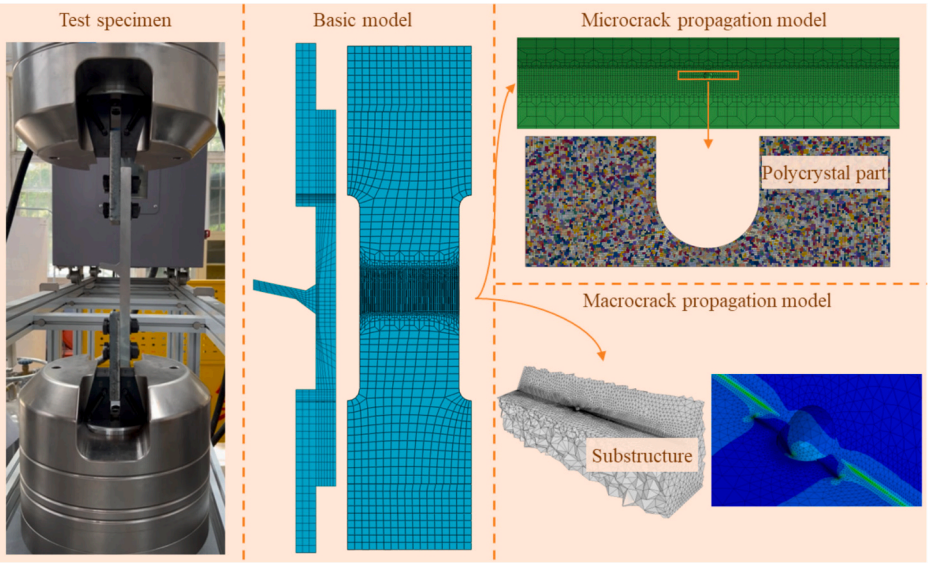


Fig. 9. Multiscale simulations of test specimens.

[36]. Therefore, this study adopts a 3D polycrystal part embedded into the macroscale model to simulate the grain-by-grain microcrack propagation process and compute the PSED values, thereby enabling a more accurate characterization of the actual microstructural features of polycrystalline materials and their mechanical responses under complex loading conditions. The crystal plasticity theory is implemented in ABAQUS through the user-defined material subroutine (UMAT) by Huang [37] to accurately calculate the anisotropic deformation behavior of materials at the crystallographic scale. The main steps are summarized as follows and illustrated in Fig. 10.

(1) Initial model establishment: Based on the test observed crack initiation locations and early propagation morphology, the polycrystal part for microcrack propagation simulation is defined as a rectangular block with dimensions of 7 × 2.5 × 0.2 mm. A cutout with the same dimensions is cut from the weld zone of the base model established in Section 4 to embed the polycrystal part. The surrounding structure outside the polycrystal part is modeled as linear elastic material with a Young's modulus of 190 GPa and a Poisson's ratio of 0.3, meshed using C3D8R elements. The boundary conditions and loading are defined according to the eccentric tensile fatigue tests, and an initial ".inp" file containing nodes, elements, materials, and boundary conditions information is generated.

(2) Polycrystal part generation and material assignment: The polycrystalline morphology and spatial grain distribution are randomly generated using the Neper based on the Laguerre-Voronoi tessellation algorithm, and a ".stess" file containing grain seed coordinates is obtained. Subsequently, a MATLAB script is employed to integrate the ".stess" file with the initial ".inp" file generated in Step (1), assign random grain orientations to each grain, and define uniform crystal plasticity material parameters. The crystal plasticity material parameters are adopted from the authors' previous study, in which they were identified using a DT framework coupled with a particle swarm optimization algorithm [38]. The polycrystal part is discretized with a mesh size of 25 μm using C3D8 elements, and the CPFEM-based microcrack propagation model is generated.

(3) Model coupling and computational preparation: The generated polycrystal part is imported into ABAQUS and coupled to the initial model using tie constraints. According to the test observed crack propagation paths, the polycrystal part is divided into two sets, namely "crystal-left" and "crystal-right," to facilitate automated crack path identification and tracking in subsequent microcrack propagation simulations.

After the model is established, the numerical simulations of microcrack initiation and propagation are conducted following the flowchart shown in Fig. 11, which consists of crack initiation analysis, single-step propagation update, and cyclic iterative computation. The detailed computational procedures are described as follows:

(1) Crack initiation analysis: After the first CPFEM simulation, post-processing is performed using a Python script developed via ABAQUS secondary development to extract the stress-strain status fields in the weld zone and compute the PSED for each grain. Automatic identification is performed for both "crystal-left" and "crystal-right," and the grain exhibiting the maximum PSED value is selected as the starting point for crack propagation. The two grains are recorded using the sets "crack-left-01" and "crack-right-01", respectively. Their crack length, depth, and corresponding PSED values are extracted as initial conditions for subsequent crack propagation analysis.

(2) Single-step propagation update: To represent the state in which the crack has already propagated and to avoid numerical non-convergence caused by element deletion, the equivalent stiffness of the cracked grain set "crack-i" is reduced to 1% of its original value. Meanwhile, the stiffness of the adjacent grain set "crack-i + 1", located ahead of the crack front grain set "crack-i", is reduced to 10% of its original value to reflect the initial damage state near the crack tip and improve computational convergence. This stiffness degradation method provides an equivalent representation of local load-carrying capacity reduction induced by crack propagation, rather than explicitly modeling geometric cracks. After stiffness degradation, the next CPFEM simulation is performed, and the updated stress-strain fields are extracted, while the grain information in "crack-i + 1" as well as the crack geometry and corresponding PSED values are automatically recorded.

(3) Cyclic iterative computation: Step (2) was repeated iteratively until the crack length reached the micro-macro transition size defined in Section 4.3, yielding the complete PSED evolution under a given random grain distribution.

Under the identical external fatigue loading conditions, including stress amplitude, frequency, and stress ratio, the crack propagation rate and propagation mode are primarily determined by the material microstructure properties [39]. To comprehensively account for the influence of randomness in grain orientation and morphology, 20

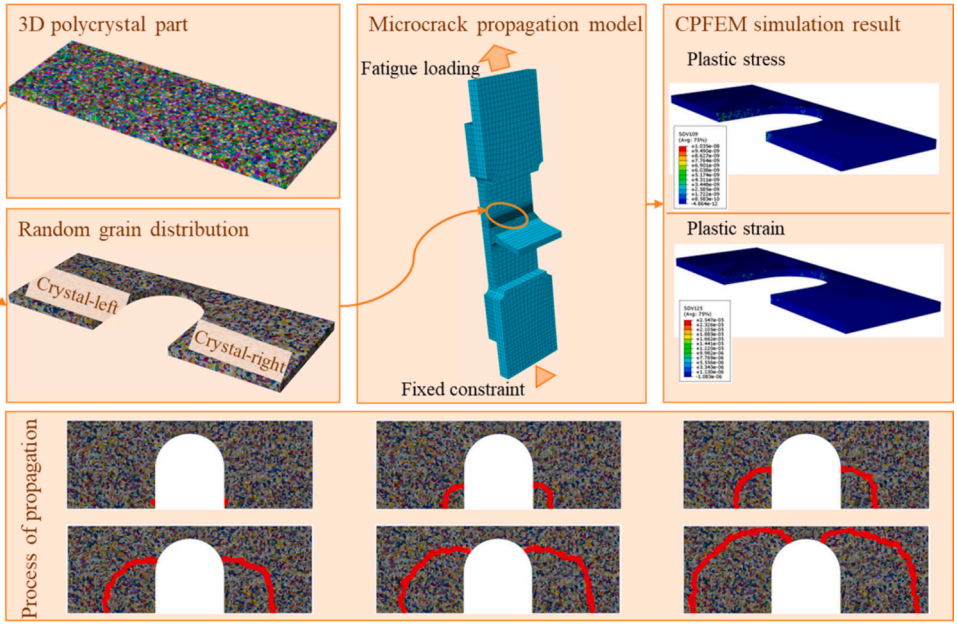


Fig. 10. Numerical simulation procedure for microcrack propagation.

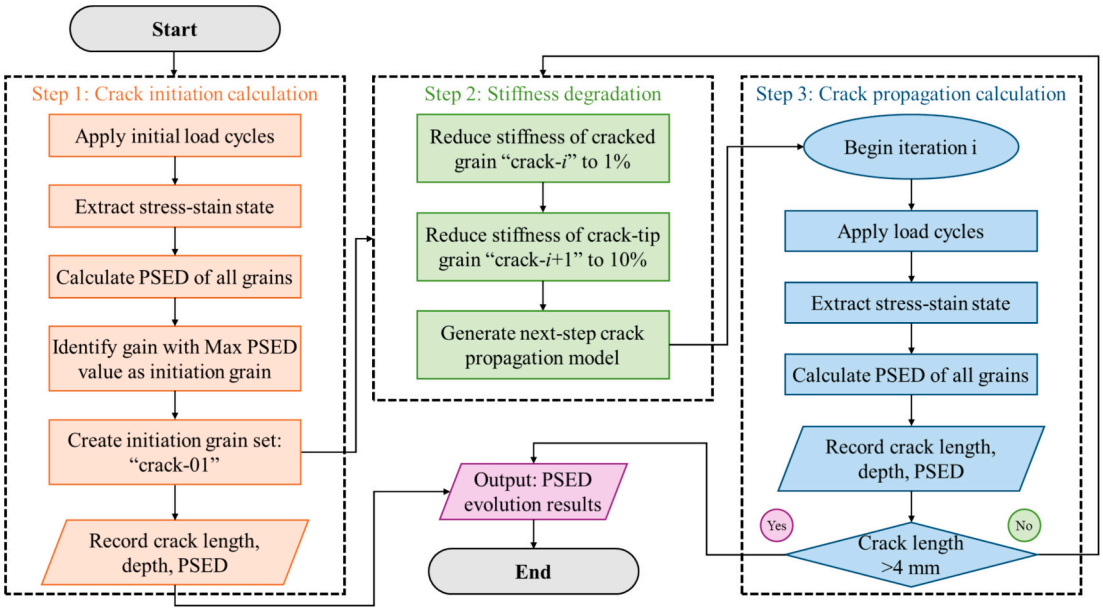


Fig. 11. Flowchart of microscopic crack propagation simulation.

CPFEM simulations with different random distributions are generated using automated scripts, and both the microcrack modeling and propagation simulation procedure described above are repeated. The resulting 20 sets of PSED evolution data are shown in Fig. 12 (yellow curves). It should be noted that the PSED obtained at the first propagation step is relatively high, as this step includes the contribution of crack initiation to fatigue life. After statistical analysis, the average PSED evolution curve (green curve) is adopted as the input for the subsequent fatigue life prediction.

4.3. Micro-macro transition

In fatigue crack propagation simulation, the transition size between microcrack and macrocrack is a key parameter that significantly influences the accuracy of fatigue life prediction. This transition size not only marks the end of the microcrack propagation stage but also represents the beginning of stable macrocrack propagation. Therefore, its determination plays a critical role in the predictive performance of the crack propagation model. It should be noted that the physical significance of this transition size is not merely determined by the artificially prescribed crack length, but it is closely associated with the microstructural characteristics of polycrystalline materials. Owing to the pronounced randomness in grain size and spatial distribution, the microcrack typically needs to traverse multiple grains before forming a macrocrack. The cumulative length of cracked grain sizes cannot be strictly associated with a fixed deterministic value. Consequently, simply defining the micro-macro transition size as a constant may fail to correspond to the actual accumulated grain sizes and may not adequately capture the actual transition from microcrack to macrocrack behavior under different microstructural, thereby potentially introducing systematic errors in fatigue life prediction.

With the continuous improvement in the numerical accuracy and stability of CPFEM-based crack propagation simulations, together with advances in computational resources, the microscale simulation of relatively long crack propagation processes has become increasingly feasible. Based on these considerations, the micro-macro transition size is treated as a random variable that varies with the underlying microstructure in this study, with its specific value determined by the cumulative size of several adjacent grains along the crack propagation path. Although the transition size corresponding to different random polycrystalline structures is not identical, its statistical distribution consistently fluctuates around a characteristic length of approximately 4 mm. The transition size of 4 mm is defined as a representative micro-macro transition length based on a comprehensive consideration of both computational efficiency and physical rationality [40]. In fatigue test, a circular hole defect with a diameter of 2 mm is introduced, from which

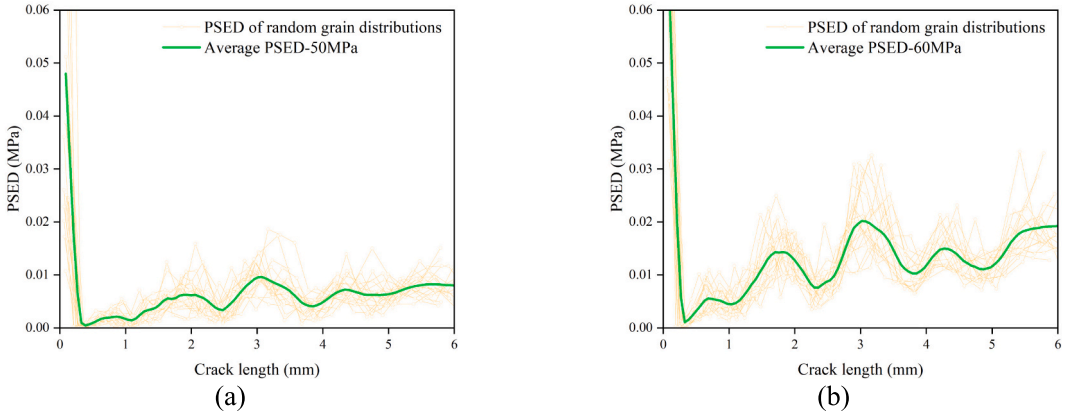


Fig. 12. PSED results: (a) 50 MPa; (b) 60 MPa.

microcracks are assumed to initiate on both sides. Each microcrack propagates independently to a maximum length of approximately 1 mm, and their coalescence results in an equivalent total crack length of about 4 mm. This transition size represents an accumulated crack length spanning multiple grains along the propagation path, rather than an individual microstructural feature. The average grain size in the HAZ is approximately 53.6 μm; thus, the length of a single microcrack (1 mm) corresponds to about 18.7 times the grain size, while the coalesced crack length (4 mm) reaches approximately 74.6 times the grain size. According to previous fatigue microcrack studies [41,42], crack propagation is strongly influenced by microstructural features when the crack length is below 10 times the grain size, while this influence gradually diminishes between 10 and 20 times, and becomes negligible beyond 20 times the grain size. Therefore, the selected transition size of 4 mm clearly falls within the macro-mechanics-dominated regime. Furthermore, to account for the inherent variability of the transition process, the transition size is treated as a stochastic variable. Taking 4 mm as the mean value, 1000 random transition lengths are generated through statistical sampling to reflect the randomness associated with different crack propagation paths.

By parameterizing the micro-macro transition size as a random variable, the proposed model retains 4 mm as a representative transition size in an average sense, while reasonably accounting for the effects of variation in grain size, grain orientation, and their combinations on the crack transition behavior. This treatment enhances the reliability and generalizability of fatigue life prediction results while maintaining physical consistency. As shown in Fig. 13, these are 1000 randomly generated micro-macro transition sizes. These random micro-macro transition sizes will be used in the analysis presented in Section 5.3.

4.4. Macrocrack propagation stage

Once microcrack propagation penetrates the polycrystal part, the crack propagation behavior is no longer primarily determined by individual grain orientations or microscale slip mechanisms. Instead, it gradually transitions into a macroscale fracture process controlled by the global geometry of the component, loading conditions, and the SIF at the crack tip. At this stage, the crack can be regarded as a macroscale defect embedded in a continuous medium, and its propagation behavior can be described using LEFM.

4.4.1. Fatigue life of macrocrack stage

The Paris law [43] is adopted in this study as the analytical model for macrocrack propagation, as expressed in Eq. (4). In the macrocrack propagation stage, the Paris law is employed to quantitatively describe the relationship between the fatigue crack propagation rate and the driving force of crack tip.

$$\mathrm{d}a/\mathrm{d}N_{\mathrm{macro}} = C(\Delta K)^m \tag{4}$$

where a denotes the crack length; N_{macro} represents the number of load cycles in the macrocrack propagation stage; C and m are material parameters associated with the fatigue crack propagation behavior.

The Paris law parameters C and m were calibrated based on the present eccentric tensile welded joint specimens under different stress levels. It should be noted that these parameters are condition-dependent and are mainly applicable to the specific specimen configuration and loading conditions considered in this study. When the mean stress level, structural geometry, or welding details change, the parameters should be recalibrated using corresponding test data.

By integrating both sides of Eq. (4), Eq. (5) can be derived. This equation is used to calculate the remaining fatigue life associated with macrocrack propagation from the initial crack length a_{i} to the final crack length a_{c}.

$$N_{\mathrm{macro}} = \int_{a_{\mathrm{i}}}^{a_{\mathrm{c}}} \frac{da}{C(\Delta K)^m} \tag{5}$$

where a_{i} denotes the initial crack size of the macrocrack, determined by the micro-macro transition size; a_{c} represents the final crack length of the macrocrack, which is typically defined as one-half or one-third of the characteristic specimen dimension.

4.4.2. Numerical simulation based on LFEM

The key parameters C and m in the Paris law are crucial to ensuring the reliability of the macrocrack propagation model. To determine the parameters C and m, a fitting analysis is performed on the relationship between the crack propagation rate $\mathrm{d}a/\mathrm{d}N_{\mathrm{macro}}$ and the SIF range ΔK. The crack propagation rate is obtained from the test results described in Section 3.3, while the corresponding ΔK values are calculated through a coupled numerical simulation using ABAQUS and Franc3D as detailed in this section.

The flowchart for establishing the macrocrack propagation model is illustrated in Fig. 14. First, the basic model developed in Section 4 is

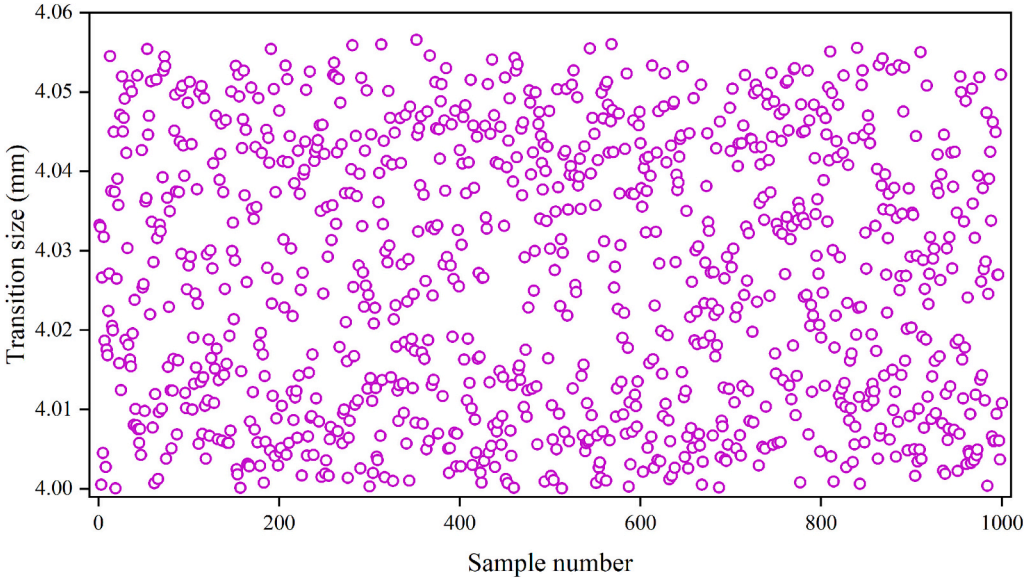


Fig. 13. Micro-macro transition sizes.

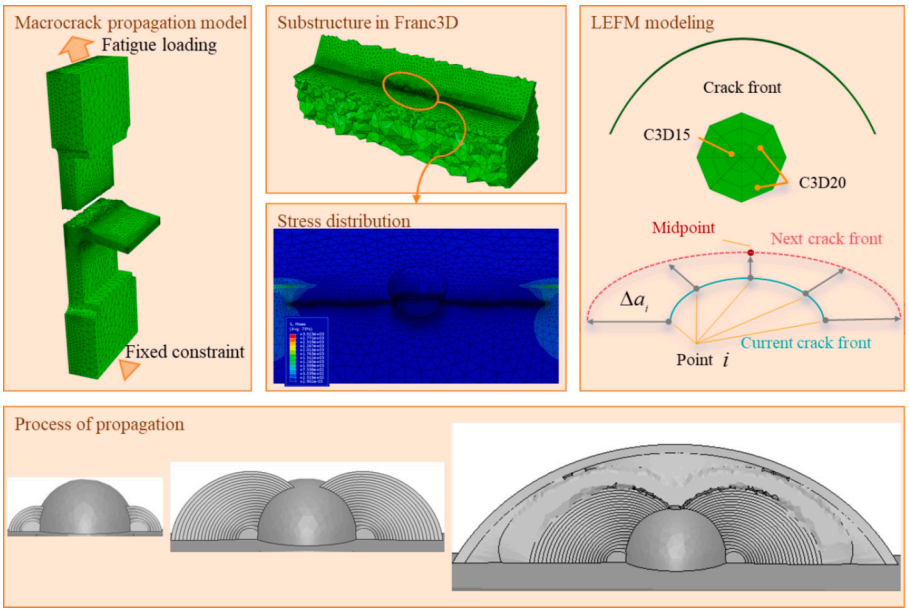


Fig. 14. Numerical simulation procedure for macrocrack propagation.

imported into Franc3D. Considering that crack propagation mainly occurs in the weld zone, only the potential macrocrack propagation zone is defined as a local substructure and discretized with a refined mesh, whereas the remaining zones are meshed more coarsely to improve computational efficiency while maintaining adequate accuracy. The global model is discretized using C3D10 elements. To ensure the numerical convergence in Franc3D calculations, two symmetric initial cracks with a length of 0.2 mm are pre-inserted at the prefabricated circular hole defect based on the crack morphologies observed in the fatigue tests. It should be noted that only the simulation results obtained after the micro-macro transition size are used in subsequent analyses. During the crack propagation simulation, Franc3D automatically generates a locally refined mesh along the semi-elliptical crack front. The crack-front region consists of three rings of elements: the middle ring is composed of C3D15 elements to accurately represent the stress singularity at the crack tip, while the two outer rings are composed of C3D20 elements to further enhance the accuracy of the crack tip field calculation. In each crack propagation step, stress analysis of the cracked model is first performed in ABAQUS to obtain the stress-strain field distribution of the structure. Subsequently, Franc3D calculates the SIF distribution along the crack front based on the obtained stress-strain field and determines the crack propagation increment and deflection angle for the next propagation step. Meanwhile, automatic remeshing is conducted in the vicinity of the crack front for the subsequent step. By repeating the above procedure, the step-by-step crack propagation path and the corresponding SIF ranges are obtained.

Numerical simulations are carried out to obtain the variation of SIFs with increasing crack length under stress amplitudes of 50 MPa and 60 MPa, respectively. To determine the SIF corresponding to each crack length observed in fatigue test, the macrocrack propagation obtained from test (i.e., after the micro-macro transition size) are combined with the numerically obtained $\Delta K - a$ relationships. An interpolation method is then employed to calculate the SIF range associated with each crack length, as shown in Fig. 15.

To obtain the parameters in the Paris model, Eq. (4) is transformed into a base-10 logarithmic form, resulting in Eq. (6). Based on Eq. (6), the parameters C and m can be determined from the linear relationship between the crack propagation rate $\mathrm{d}a/\mathrm{d}N_{\mathrm{macro}}$ and the SIF range ΔK in a double-logarithmic coordinate system. The least-squares method is then employed to fit the test data at each propagation stage, resulting in a straight fitting line, as shown in Fig. 16. The slope of this line corresponds to the parameter m, while the intercept on the Y-axis corresponds to $\lg C$, from which the parameters C and m can be determined. These parameters will be used in the analysis presented in Section 5.4.

$$\lg(\mathrm{d}a/\mathrm{d}N_{\mathrm{macro}}) = \lg C + m\lg(\Delta K) \tag{6}$$

5. Data fusion and digital twin

In DT, both physical entity and virtual entity are inevitably affected

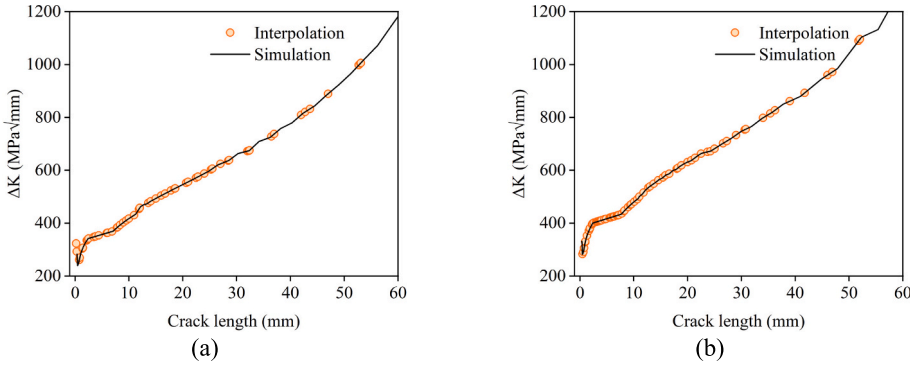


Fig. 15. Simulation results of SIF range vs crack length: (a)50 MPa; (b)60 MPa.

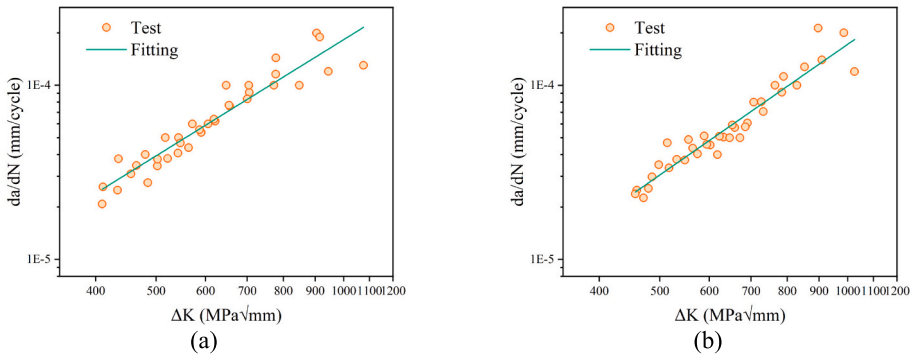


Fig. 16. Fitting process of Paris model: (a) 50 MPa; (b) 60 MPa.

by multi-source uncertainties, which degrade the predictive accuracy of the DT. Consequently, data fusion, as a core component in the establishment of DT, plays a crucial role in mitigating these adverse effects.

The uncertainties associated with physical entity mainly originate from two aspects. One aspect is that the manufacturing process itself is inherently stochastic. During the specimen fabrication process, tolerances inherent in the cutting procedure and the drilling of a prefabricated circular hole led to a dispersed distribution of initial crack morphologies and sizes. Another aspect is that limitations in sensor monitoring capabilities also constitute a significant source of uncertainty. Owing to environmental noise and sensor accuracy constraints, measurement errors are inevitably embedded in the monitoring data. In contrast, the uncertainties of the virtual entity primarily come from model systematic deviations. Numerical simulation models are essentially idealized and simplified representations of real physical processes, and various approximations and assumptions are inevitably introduced during model construction. These include stochastic deviations in microstructure generation algorithms, idealized assumptions in boundary condition, and simplified treatments of key parameters such as the SIF during numerical simulation calculation. Collectively, these factors lead to discrepancies between the virtual entity responses and the actual physical entity behavior.

The superposition of these multi-source uncertainties can significantly compromise the accuracy of crack propagation simulations. Therefore, it is necessary to employ data fusion techniques to comprehensively integrate information from different sources and at different levels and to achieve effective coupling between physical entity data and virtual entity data through appropriate algorithms, thereby alleviating the influence of uncertainties on the DT establishment process [44]. Through data fusion, the DT can more accurately reflect the actual state of the physical entity during its dynamic evolution, enhancing both the accuracy and credibility of its predictive results and providing a more reliable basis for engineering assessment and decision-making. Accordingly, this study introduces a DBN framework in combination with a PF algorithm to perform dynamic updating of the crack propagation process across its two distinct propagation stages.

5.1. Dynamic Bayesian networks

DBN is a class of probabilistic graphical models designed to characterize stochastic processes evolving over time. Owing to its strong capability in handling nonlinear and non-Gaussian uncertainties, as well as its support for online updating of both model parameters and system states, DBN has been widely applied in DT framework [45]. Within the DBN framework, heterogeneous information derived from fatigue tests and numerical simulations can be effectively fused. This fusion enables real-time correction of both model parameters and state estimates as new observational data become available. Consequently, the crack propagation states can be continuously updated and predicted, thereby significantly improving the accuracy of fatigue life prediction [46].

Essentially, DBN is constructed by unfolding a static Bayesian network along the temporal dimension to form a sequential probabilistic model. Its fundamental assumptions include first-order Markov property and temporal stationarity, meaning that the system state at the current time depends only on the state at the previous time step and is independent of earlier historical states. Within the DBN framework, the crack propagation process can be abstracted as a stochastic dynamic system in which the hidden state evolves over time. Specifically, the crack state variables are defined as latent states to describe the evolution behavior of cracks under cyclic loading, while the corresponding test measurements are treated as observation variables that provide constraints for the estimation and correction of the latent states. By establishing a state transition model and an observation model, the DBN can systematically describe the temporal dependency between hidden crack states and observational information during the crack propagation process. The overall schematic structure of the DBN employed in this study is illustrated in Fig. 17.

After discretizing time, the hidden state of the system at time instant t is denoted as X_t, and the corresponding observation data are denoted as Y_t. DBN characterizes the system evolution mechanism through the probabilistic descriptions of the state transition model and the observation model. Accordingly, the joint probability distribution of the hidden states and observations can be factorized as

$$p(X_{1:T}, Y_{1:T}) = p(X_1)p(Y_1|X_1)\prod_{t=2}^{T} p(X_t|X_{t-1})p(Y_t|X_t) \tag{7}$$

where $p(X_t|X_{t-1})$ represents the state transition model, which is used to describe the evolution law of the crack propagation state between two adjacent time steps. The hidden state X_t characterizes the system state at time t and generally consists of a crack propagation model combined with state noise, as expressed in Eq. (8), to account for the evolution of fatigue damage and its inherent uncertainty, and $p(X_1)$ denotes the initial state distribution of the system. The conditional probability $p(Y_t|X_t)$ corresponds to the observation model, which establishes the mapping relationship between the hidden state and the sensor measurement data as well as the numerical simulation results. The observation Y_t denotes the measurement data at time t, whose formulation is given in Eq. (9). It is typically composed of the actual observation and measurement noise, thereby reflecting the influence of observation errors on the crack propagation state.

$$X_t = T(X_{t-1}, \eta_t) \tag{8}$$

$$Y_t = O(Y_{t-1}, \varepsilon_t) \tag{9}$$

where T denotes the state transition function; O denotes the observation function; η_t and ε_t represent the state noise and observation noise, respectively, which are introduced to characterize the uncertainties in the system evolution and observation process.

Given all observation data up to time t, denoted as $Y_{1:t} =$

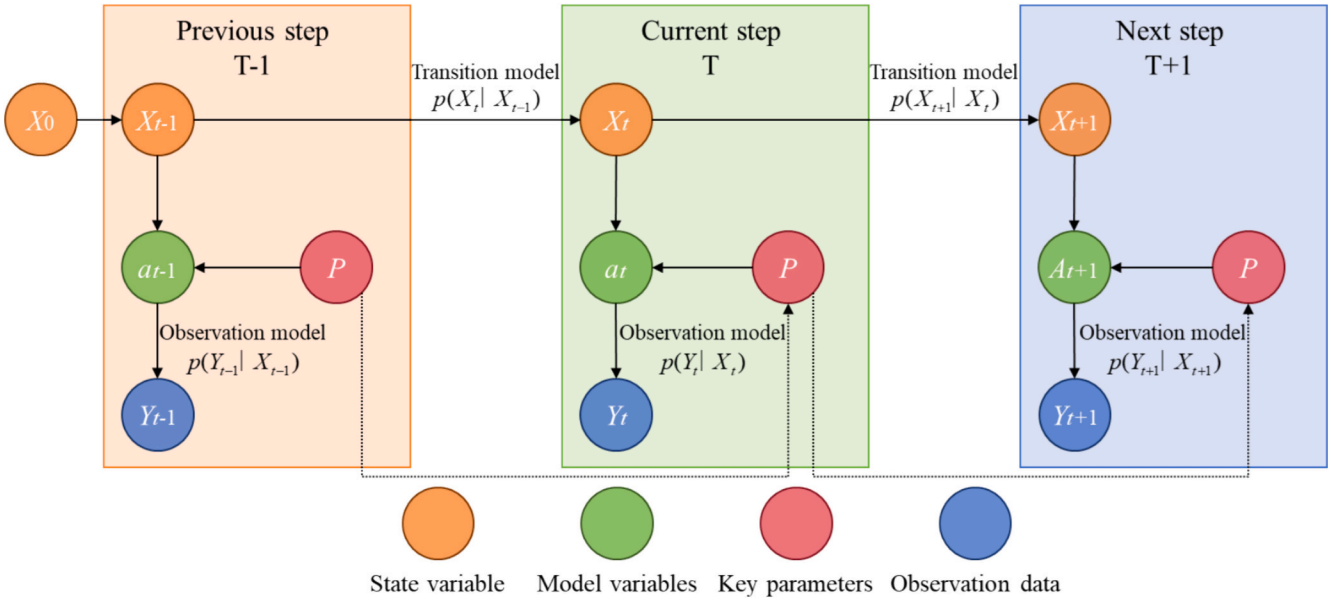


Fig. 17. Dynamic Bayesian network framework.

$\{Y_1, Y_2, \ldots, Y_t\}$, the DBN is capable of inferring the posterior probability distribution of the current hidden state $p(X_t|Y_{1:t})$. Based on recursive Bayesian inference, continuous prediction and dynamic updating of the crack propagation state can be achieved. In this study, different state transition models are employed for the microcrack propagation stage and the macrocrack propagation stage, respectively. Specifically, the microcrack propagation stage is based on the PSED-based fatigue life calculation formula, whereas the macrocrack propagation stage is based on the Paris crack propagation law. For both stages, the crack length is selected as the observation variable in the observation model. Meanwhile, the micro-macro transition size delineating the microcrack and macrocrack regimes is adopted as the criterion for stage switching. Through this way, continuous simulation and dynamic updating of the multiscale crack propagation model are realized within a unified DBN framework.

It should be noted that the first-order Markov property and temporal stationarity adopted in the DBN model are engineering approximations rather than strict physical representations of fatigue crack propagation. Although fatigue crack propagation may exhibit path-dependent behavior, the current crack state is considered to implicitly contain the cumulative effects of prior loading history, making the first-order Markov assumption a practical simplification. The temporal stationarity assumption is mainly applicable to conditions involving piecewise constant loading or slowly varying statistical characteristics. In addition, the observation-based sequential updating mechanism incorporated in the model allows parameters to be continuously updated as new data become available, which partially accounts for historical effects and improves the adaptability of the model.

5.2. PF algorithm

Since the hidden states in DBN evolve dynamically over time and are updated in a recursive manner, their posterior distributions can only be progressively refined based on the currently available observations. Consequently, during the iterative processes of state estimation and parameter updating, it is necessary to introduce efficient filtering algorithms to achieve online fusion of historical information, thereby avoiding explicit enumeration of the complete state sequence and effectively reducing computational complexity while improving inference efficiency. Among various filtering algorithms, the PF algorithm has been widely adopted within the DBN framework for state inference and updating problems [47]. This is primarily attributed to its strong capability to approximate posterior distributions in nonlinear and non-Gaussian systems, its favorable characteristics for parallel computation, and its ability to simultaneously perform joint estimation of system states and model parameters. The PF algorithm is a Bayesian inference method based on the sequential Monte Carlo, in which the posterior probability distribution of the system state is numerically approximated by a set of weighted particles. In the PF algorithm, recursive state estimation is realized through successive steps of particle propagation, weight updating, and resampling. In this study, each randomly generated particle corresponds to a reasonable combination of key parameters in the crack propagation model, and the entire particle set statistically represents the probability distributions of these key parameters and state variables. At each iteration, the PF algorithm directly approximates posterior probability distribution using Monte Carlo sampling techniques. As new monitoring data are continuously incorporated, the particle set can be resampled and reweighted, enabling synchronous updating and real-time prediction of the fatigue crack propagation

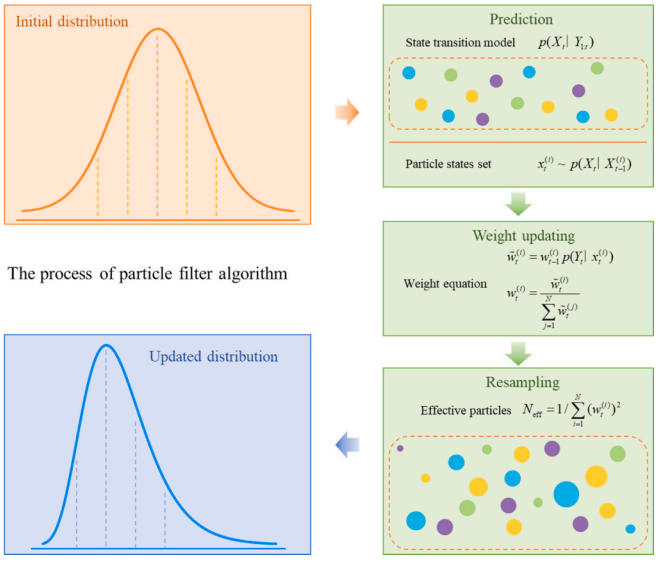


Fig. 18. Flowchart of the PF algorithm.

process. The flowchart of the PF algorithm is shown in Fig. 18.

The PF algorithm approximates the posterior distribution of the system state X_t using a set of weighted particles $\left\{\left\{x_t^{(i)}, w_t^{(i)}\right\}_{i=1}^{N}\right\}$, which can be expressed by Eq. (10):

$$p(X_t|Y_{1:t}) \approx \sum_{i=1}^{N} w_t^{(i)} \delta\left(x_t - x_t^{(i)}\right) \tag{10}$$

where $w_t^{(i)}$ denotes the weight of the i-th particle; and $\delta()$ is the Dirac delta function.

By continuously incorporating new observation data and updating the particle set, the PF algorithm is capable of efficiently estimating the posterior distribution of hidden states in complex time-series systems and gradually approximating the actual distribution. As time evolves, the amount of fused information increases, leading to improved accuracy and stability of the system state estimation. At each time step, the PF algorithm mainly consists of three steps: prediction, update, and resampling, which are described as follows:

(1) Prediction: Based on the particle set at the previous time step, the state transition model of the DBN is used to propagate each particle and generate a set of candidate particles for the current time step:

$$x_t^{(i)} \sim p\left(X_t|X_{t-1}^{(i)}\right) \tag{11}$$

As a result, a Monte Carlo approximation of the predictive distribution $p(X_t|Y_{1:t-1})$ is obtained as:

$$p(X_t|Y_{1:t-1}) \approx \sum_{i=1}^{N} w_{t-1}^{(i)} \delta\left(x_t - x_t^{(i)}\right) \tag{12}$$

(2) Update: Sequential importance sampling is employed to update the particle weights according to the observation data at the current time step to reflect the ability of each particle to explain the observations:

$$\widetilde{w}_t^{(i)} = w_{t-1}^{(i)} p\left(Y_t|x_t^{(i)}\right) \tag{13}$$

The weights are then normalized to ensure that their sum equals unity, thereby avoiding numerical divergence and allowing the relative importance of particles to be correctly represented:

$$w_t^{(i)} = \frac{\widetilde{w}_t^{(i)}}{\sum_{j=1}^{N} \widetilde{w}_t^{(j)}} \tag{14}$$

(3) Resampling: As time progresses, particle weights tend to concentrate on a small number of particles, while the weights of most particles approach zero, resulting in particle degeneracy and reducing the ability of the particle set to effectively represent the actual posterior distribution. To alleviate this problem, the effective sample size N_{eff} is introduced:

$$N_{\text{eff}} = \frac{1}{\sum_{i=1}^{N} \left(w_t^{(i)}\right)^2} \tag{15}$$

When N_{eff} falls below a prescribed threshold N_{thr}, a resampling operation is performed to generate a new set of equally weighted particles:

$$w_t^{(i)} = \frac{1}{N} \tag{16}$$

5.3. Data fusion for microcrack propagation

The observation equation for the microcrack propagation stage is established based on test data spanning from the 0 mm to the micro-macro transition size. In this study, the relationship between crack length and loading cycles obtained from a single specimen is selected as the observation to characterize the propagation behavior of microcrack propagation.

At this stage, crack propagation is controlled by the combined effects of material microstructural characteristics and local mechanical responses. Based on Eq.(3), the microcrack propagation model is discretized, and a stochastic perturbation term is introduced to account for the uncertainties induced by material heterogeneity and model simplification. Accordingly, the state transition equation for the microcrack propagation stage can be expressed as follows:

$$X_t = X_{t-1} + r_i \Delta N \frac{\pi E f^* (\varpi + \eta_{t-1})}{4\pi E \gamma_s - 4\sigma^2 d_t (1-\nu^2)} \tag{17}$$

where X_t and X_{t-1} denote the crack state variables at the current and previous time steps, respectively. The difference between the two values is defined as the crack increment Δa. The crack increment is calculated as the crack propagation rate per cycle multiplied by ΔN, where ΔN denotes the fixed number of loading cycles between two adjacent time steps; r_i is a dimensionless random grain size factor, defined as $r_i = d_i/\overline{d}$;η_{t-1} represents the noise term introduced to describe the random perturbations arising from material microstructural inhomogeneity and model simplifications during microcrack propagation.

The parameter f^* is an updating model parameter, whose initial uncertainty is determined based on fatigue test results. Considering the discreteness of parameter values, a lognormal distribution is adopted to characterize its uncertainty. Specifically, the 50 MPa condition satisfies$\ln f^* \sim N(-14.82, 0.995^2)$, while the 60 MPa condition satisfies$\ln f^* \sim N(-16.61, 0.995^2)$.

5.4. Data fusion for macrocrack propagation

The observation equation for the macrocrack propagation stage is established based on test data spanning from the micro-macro transition size to the final fracture stage of the specimen. Similarly, the relationship between crack length and loading cycles obtained from a single specimen is employed as the observation information to describe the evolution law of macrocrack propagation.

Once the crack enters the macrocrack propagation stage, its propagation behavior is governed by the SIF range and can be described using the Paris law. Based on Eq. (4), and taking into account the stochastic uncertainties involved in the crack propagation process, the state transition equation for the macrocrack propagation stage is formulated as:

$$X_t = X_{t-1} + \Delta N C (\Delta K)^m + \eta_{t-1} \tag{18}$$

where C and m are selected as the updating parameters [48]. Their initial uncertainties are determined based on test calibration results, with C characterized by lognormal prior distributions and m by normal prior distributions. Specifically, the 50 MPa condition satisfies$\ln C \sim N(-24.045, 0.25^2)$, $m \sim N(2.2168, 0.25)$, while the 60 MPa condition satisfies$\ln C \sim N(-26.01, 0.25^2)$, $m \sim N(2.4928, 0.25)$.

6. Applications and predictions

During the crack propagation prediction process, the number of particles in the PF is set to 1000. This particle scale provides a favorable balance between computational efficiency and prediction accuracy, and preliminary numerical trials confirm that it yields stable and reliable posterior distributions. On this basis, the crack propagation process under nominal stress amplitudes of 50 MPa and 60 MPa are predicted, respectively.

Fig. 19 presents the predicted crack propagation curve of the eccentric tensile specimens and compares the prediction distributions with the corresponding test observations. It depicts the DBN inference and parameter updating process in detail. The light purple vertical lines

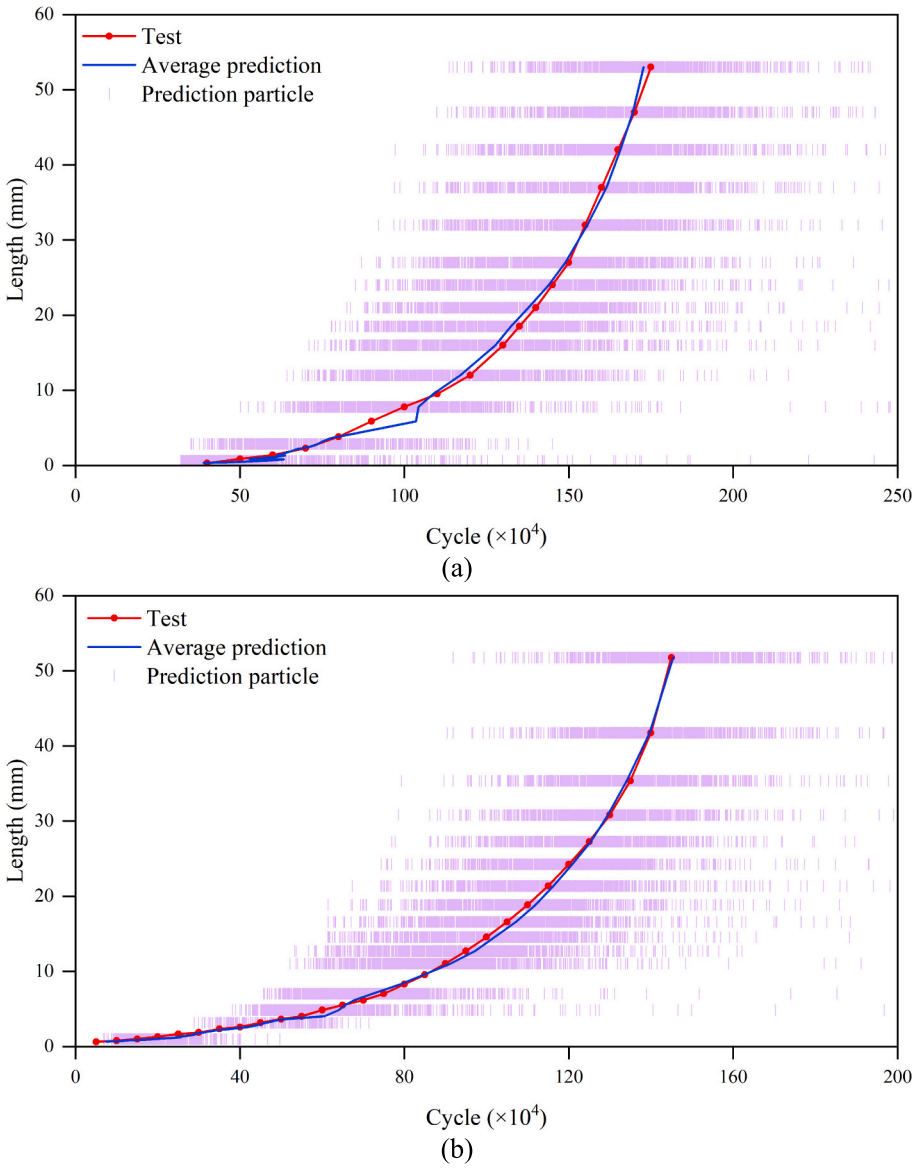


Fig. 19. Prediction of crack propagation curve: (a) 50 MPa; (b) 60 MPa.

represent the fatigue life values predicted by each particle in the PF. At each fusion point, a total of 1000 PF particles are used, resulting in 1000 predicted fatigue life values. The blue curve denotes the average prediction fatigue life calculated from the updated parameter samples, i.e., the average value of the 1000 PF particle predictions at each fusion point. Taking the 50 MPa loading condition as an example, 4 crack length locations are selected for DBN inference in the microcrack propagation stage based on the test measurements. The model calibration was conducted at crack lengths of 0.8484, 1.3718, 2.2804 and 3.8177 mm. Similarly, in the macrocrack propagation stage, 14 crack length locations are selected for DBN inference. The model calibration is performed at crack lengths of 5.88, 7.79, 9.5, 12, 16, 18.5, 21, 24, 27, 32, 37, 42 47 and 53 mm. In the microcrack propagation stage, from the initial crack length to the micro-macro transition size, the initial predictions exhibit noticeable discrepancies from the test observations due to the relatively large prior uncertainty of the key model parameters in this stage. With the progressive updating of the DT, the prediction results gradually converge after fusing the observational data through 4 DBN inference updates, showing a high level of consistency with the test measured crack propagation life. When the crack length exceeds the micro-macro transition size and enters the macrocrack propagation stage, significant deviations between the theoretical predictions based on the initial parameters and the test results are also observed, mainly due to the large uncertainty associated with the Paris law parameters C and m. By fusing multiple sets of observational data and performing successive parameter updates, the prediction accuracy is substantially improved, and the predicted fatigue life shows good agreement with the test results.

Fig. 20 illustrates the predicted number of cycles for the propagation of the last grain during the microcrack stage, obtained using different micro-macro transition sizes randomly generated in Section 4.3. The results show that incorporating a micro-macro transition size does not materially affect the predicted cycle numbers, thereby confirming that its introduction does not compromise the overall accuracy of the model. At the same time, this treatment enables the stochastic nature of grain propagation to be more realistically represented within the framework. Therefore, the introduction of the micro-macro transition size is both physically meaningful and methodologically justified in the present modeling approach.

Fig. 21 further illustrates the probability density function of the predicted fatigue life at different test observation points. The probability density curves exhibit a systematic rightward shift with increasing crack

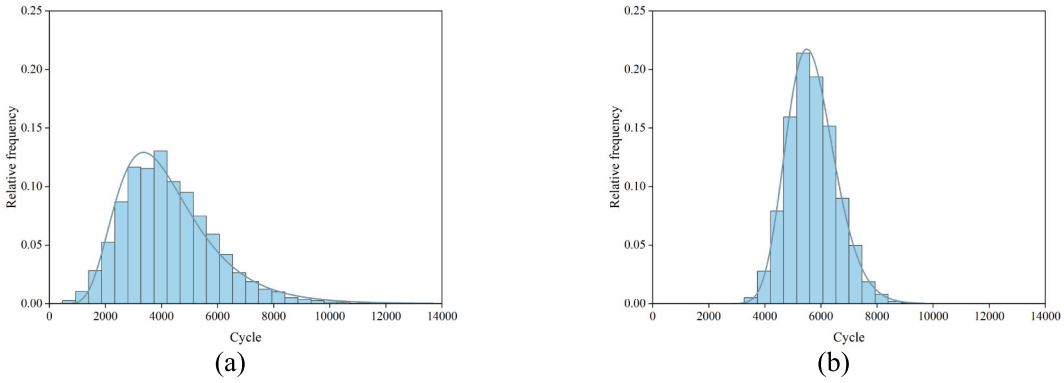


Fig. 20. Cycle numbers at the micro-macro transition size: (a) 50 MPa; (b) 60 MPa.

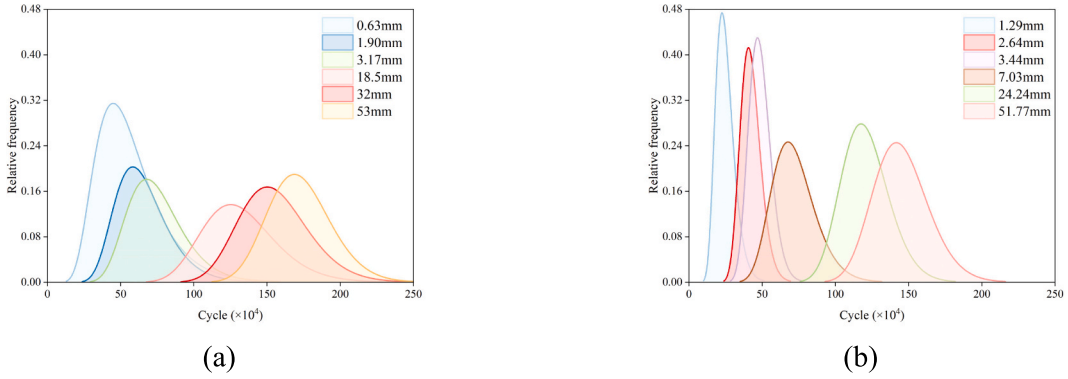


Fig. 21. Probability density distribution of fatigue life: (a) 50 MPa; (b) 60 MPa.

length, indicating a gradual increase in the predicted fatigue life. At the same time, the peak values of the distributions tend to decrease and the curves become broader as the crack length increases, reflecting enhanced dispersion and uncertainty in the predicted results at locations farther from the critical region. In addition, when the stress level increases from 50 MPa to 60 MPa, the distributions as a whole move toward lower cycle ranges, demonstrating a general reduction in fatigue life under higher stress. Therefore, regardless of the stress amplitude, the predicted fatigue life shows consistent spatial evolution characteristics, while the stress level primarily affects the overall magnitude of the life distribution.

To quantitatively evaluate the improvement in prediction performance during the iterative updating process, a prediction error indicator is introduced and defined as Eq. (19):

$$E = \frac{|P - T|}{T} \times 100\% \tag{19}$$

where E denotes the relative prediction error; P is the predicted fatigue life; T the fatigue life measured in the test.

Table 5 and Table 6 summarize the fatigue life prediction errors at different stages of the DBN updating process. Taking the 50 MPa loading condition as an example, the results indicate that for microcrack propagation, the relative prediction error is approximately 15.88% when the model is not updated. After the first model update, the relative prediction error decreases to approximately 4.34%. With the continued propagation of the fatigue crack and the progressive fusion of observational data, the relative prediction error is reduced to approximately −0.30% after the fourth update, indicating a substantial improvement in prediction accuracy. Owing to the effect of the micro-macro transition sizes, the prediction error re-increases upon entering the macrocrack stage after the end of the microcrack stage. For macrocrack propagation, the relative prediction error is approximately 56.61% before model updating and decreases to approximately 9.16% after the first update. After 6 successive updates, the relative prediction error is further reduced to approximately −0.10%, demonstrating a significant enhancement in prediction accuracy. It can be observed that when only a single observation is incorporated, the predicted mean life deviates significantly from the true fatigue life. As more test data are successively fused, the prediction accuracy continuously improves with increasing calibration iterations. It demonstrates the continuous enhancement of model prediction accuracy.

Table 5
Iterative update error of 50 MPa.

Fusion point	Propagation stage	Predicted life	Actual life	Relative error (%)
1	Microcrack	579,381.9	500,000	15.88
2		626,032.5	600,000	4.34
3		689,638.6	700,000	−1.48
4		797,599.9	800,000	−0.30
1	Macrocrack	1,400,519	900,000	55.61
2		1,091,565	1,000,000	9.16
4		1,159,026	1,200,000	−3.41
5		1,277,188	1,300,000	−1.76
7		1,371,069	1,400,000	−2.07
8		1,428,188	1,450,000	−1.50
10		1,584,433	1,550,000	2.22
11		1,610,116	1,600,000	0.63
13		1,701,736	1,700,000	0.10

Fig. 22 and Fig. 23 illustrates the updating results of the key model parameters in the microcrack and macrocrack propagation stages. In the microcrack propagation stage, the variance of the posterior distributions of the model parameters f^* continuously decreases with increasing update iterations, indicating that the parameter uncertainty is effectively suppressed through the incorporation of observational data. In the

Table 6
Iterative update error of 60 MPa.

Fusion point	Propagation stage	Predicted life	Actual life	Relative error (%)
1		147,282	100,000	47.28
3		272,325	200,000	36.16
5	Microcrack	308,544	300,000	2.85
7		418,385	400,000	4.6
9		513,765	500,000	2.75
1		666,282.9	550,000	21.14
2		674,901.6	600,000	12.48
4		686,278.3	700,000	−1.96
5		728,160	750,000	−2.91
7		833,819.2	850,000	−1.90
8		918,268.1	900,000	2.03
10	Macrocrack	1,018,908	1,000,000	1.89
11		1,074,196	1,050,000	2.30
13		1,169,182	1,150,000	1.67
14		1,215,744	1,200,000	1.31
16		1,302,030	1,300,000	0.16
17		1,358,014	1,350,000	0.59
18		1,412,214	1,400,000	0.87

macrocrack propagation stage, the large initial uncertainty of the Paris law parameters C and m leads to significant discrepancies between the predicted and test results. As additional observational data are progressively incorporated, the posterior distributions of parameters C and m gradually converge, and their uncertainties are substantially reduced.

7. Conclusions

Overall, the digital twin-based life prediction framework for weld joints developed in this study effectively integrates test data through DBN, thereby reducing the uncertainties inherent in fatigue crack propagation models. By enabling real-time calibration and continuous updating of crack propagation behavior and remaining fatigue life, the proposed framework progressively improves prediction accuracy. This capability facilitates effective structural health management of crack-prone welded details and ensures that the structure operates within a safe service range. The main conclusions from this study can be summarized as follows:

(1) A digital twin-based fatigue crack prediction framework tailored for OSD weld joints is proposed, which realizes unified modeling and description of microcrack initiation, microcrack propagation, and macrocrack propagation throughout the entire fatigue process.

(2) A multiscale crack propagation model coupling CPFEM with fracture mechanics is established as a virtual entity, which simultaneously accounts for microstructural mechanisms and macroscale fracture behavior, thereby improving the accuracy of crack propagation simulation.

(3) By integrating a DBN with a PF algorithm, real-time fusion and updating of test data and numerical simulation models are achieved and the DT is established, effectively reducing model parameter uncertainty and enhancing the reliability of fatigue life prediction.

(4) Prediction validation demonstrates that the proposed DT can accurately predict the crack propagation behavior and remaining fatigue life of weld joints, providing a sound scientific basis for health assessment and maintenance decision-making of steel bridge.

CRediT authorship contribution statement

Yang Ye: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **You-Lin Xu:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Wulve Lao:** Writing – original draft, Investigation, Formal analysis, Data

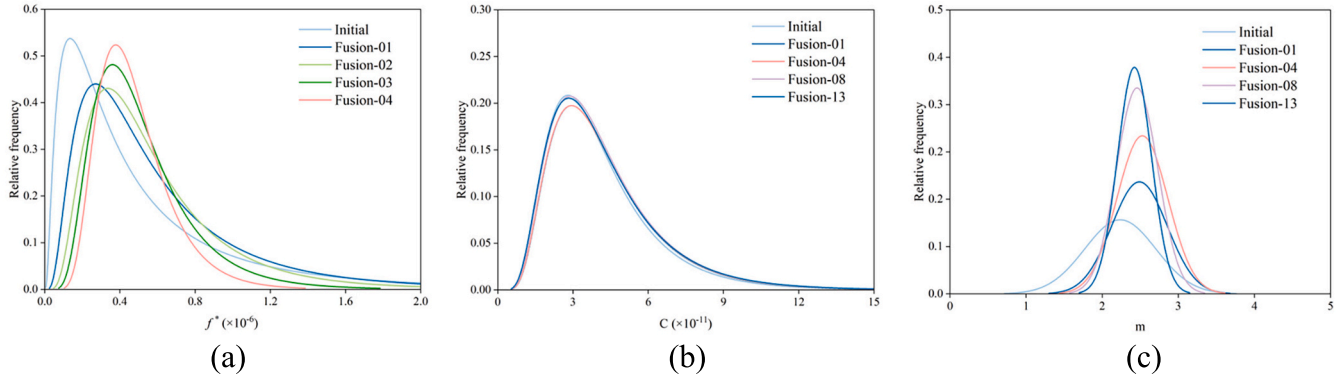


Fig. 22. Evolution of key parameter distributions of 50 MPa: (a) f^*; (b) C; (c) m.

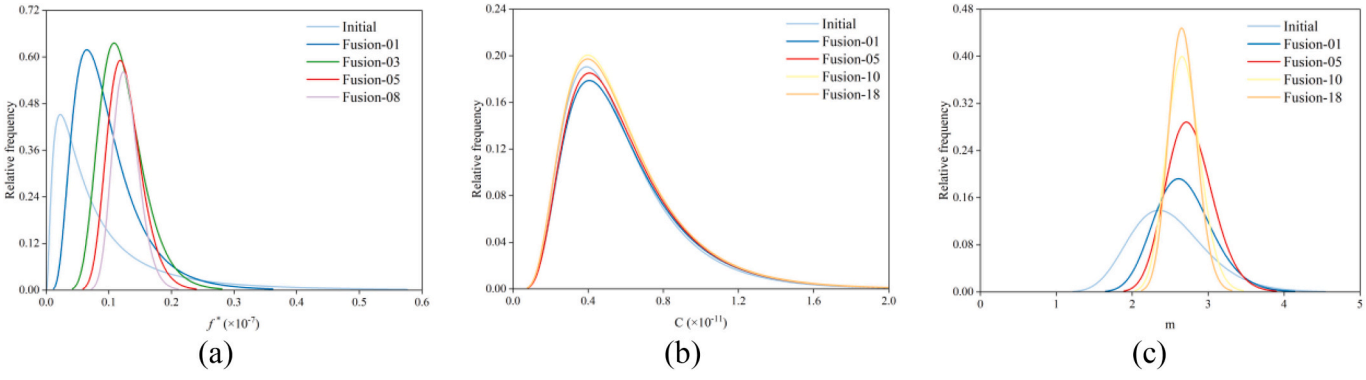


Fig. 23. Evolution of key parameter distributions of 60 MPa: (a) f^*; (b) C; (c) m.

curation. **Chuang Cui:** Writing – review & editing, Methodology, Investigation. **Qinghua Zhang:** Project administration, Methodology.

Declaration of competing interest

The authors declare that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data, models, or codes that support the findings of this study are available upon reasonable request.

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